Jeremy Tan

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Education

Imperial College London — London, UK
MEng Electronic & Information Engineering — *Oct 2023 – Jun 2027*

- Third Year: 81.5%, Dean's List (Top 5% of the year)
- Master's Thesis: *Tiny Concept-PFNs: Compact Bayesian Inference for Reliable Decisions*, supervised by Dr Sonali Parbhoo

Experience

Digital Verification Intern — Apr 2026 – Sep 2026
Qualcomm — *Cambridge, UK*

- Built an MCP (Model Context Protocol) server that lets LLM agents query binary waveform databases directly, speeding up hardware debug.
- Piloted PyUVM and cocotb, Python test frameworks for hardware, to reduce testbench development time.
- Verified low power radio subsystems for wearable chips using UVM.

Undergraduate Research Assistant — Jul 2024 – Sep 2024
Imperial College London — *London, UK*

- Assisted development of LQR, MRAC, and neural network quadcopter controllers in MATLAB/Simulink, integrated with ROS and Gazebo for simulation testing.

Projects

C to RISC-V Compiler | *C++, Flex, Bison* — Oct 2024 – Mar 2025

- Built a compiler from preprocessed C90 to RISC-V assembly in C++ with Flex and Bison, covering lexing, parsing, semantic analysis, and code generation.
- Ranked in the top decile of the cohort; passed all public tests and 80% of unseen tests.

PintOS (Toy Operating System) | *C, QEMU, GDB* — Jul 2025 – Sep 2025

- Implemented an MLFQS scheduler and user process support: stack argument passing, 13 system calls covering process control and file I/O, and user pointer validation that kills faulting processes without leaking resources.

FPGA Accelerated Fluid Simulation | *Python, C++, CUDA, SystemVerilog* — May 2025 – Jun 2025

- Implemented the Lattice Boltzmann Method in Python, C++, CUDA, and SystemVerilog, benchmarking performance across CPU, GPU, and FPGA.
- Won **Best Second Year Group Project**

Piano Synthesiser | *C++, FreeRTOS, STM32* — Jan 2026 – Mar 2026

- Built a polyphonic piano synthesiser with FreeRTOS task scheduling verified by rate monotonic analysis, and CAN bus messaging between boards.

Twitter Clone | *React, Express, Node.js, PostgreSQL* — Aug 2024 – Sep 2024

- Built a full stack social media app with authentication, profiles, follow requests, posts, comments, likes, and a feed of posts from followed users.

Awards

Dean's List, Imperial College London (2026) | Best Second Year Group Project (2025) | Bending Spoons First Commit Finalist (2026)

Technical Skills

Languages: Python, C, C++, TypeScript/JavaScript, SQL, CUDA, SystemVerilog
Frameworks: React, Node.js, Express, FastAPI, PyTorch, NumPy, cocotb, FreeRTOS
Tools: Git, Docker, Jenkins, PostgreSQL, GDB, QEMU, Verilator, Vivado
Spoken Languages: English, Mandarin, Malay